

Introduction to Python

Run the hidden code cell below to import the data used in this course.

```
# Importing course packages; you can add more too!
import numpy as np
import math

# Import columns as numpy arrays
baseball_names = np.genfromtxt(
    fname="baseball.csv", # This is the filename
    delimiter=",", # The file is comma-separated
    usecols=0, # Use the first column
    skip_header=1, # Skip the first line
    dtype=str, # This column contains strings
)
baseball_heights = np.genfromtxt(
    fname="baseball.csv", delimiter=",", usecols=3, skip_header=1
)
baseball_weights = np.genfromtxt(
    fname="baseball.csv", delimiter=",", usecols=4, skip_header=1
)
baseball_ages = np.genfromtxt(
    fname="baseball.csv", delimiter=",", usecols=5, skip_header=1
)

soccer_names = np.genfromtxt(
    fname="soccer.csv",
    delimiter=",",
    usecols=1,
    skip_header=1,
    dtype=str,
    encoding="utf",
)
soccer_ratings = np.genfromtxt(
    fname="soccer.csv",
    delimiter=",",
    usecols=2,
    skip_header=1,
    encoding="utf",
)
soccer_positions = np.genfromtxt(
    fname="soccer.csv",
    delimiter=",",
    usecols=3,
    skip_header=1,
    encoding="utf",
    dtype=str,
)
soccer_heights = np.genfromtxt(
    fname="soccer.csv",
```

```
    delimiter="," ,
    usecols=4,
    skip_header=1,
    encoding="utf",
)
soccer_shooting = np.genfromtxt(
    fname="soccer.csv",
    delimiter="," ,
    usecols=8,
    skip_header=1,
    encoding="utf",
)
```

Take Notes

Add notes about the concepts you've learned and code cells with code you want to keep.

Add your notes here

```
# Add your code snippets here
hall = 12
kit = 14
bed = 24
bath = 44
areas= [{"hallways", hall}, {"kitchen", kit}, {"bedroom", bed}, {"bathroom", bath}]
print(areas)

[['hallways', 12], ['kitchen', 14], ['bedroom', 24], ['bathroom', 44]]
```

Find in the hidden cell below some exercises to explore the data and practice your skills:

Explore Datasets

Use the arrays imported in the first cell to explore the data and practice your skills!

- Print out the weight of the first ten baseball players.
- What is the median weight of all baseball players in the data?
- Print out the names of all players with a height greater than 80 (heights are in inches).
- Who is taller on average? Baseball players or soccer players? Keep in mind that baseball heights are stored in inches!
- The values in `soccer_shooting` are decimals. Convert them to whole numbers (e.g., 0.98 becomes 98).
- Do taller players get higher ratings? Calculate the correlation between `soccer_ratings` and `soccer_heights` to find out!
- What is the average rating for attacking players (`'A'`)?

```
import numpy as np

# In a real scenario, these arrays would be loaded from a dataset

# Baseball data
baseball_weights = np.array([180, 175, 200, 190, 185, 210, 195, 170, 205, 188, 192,
198, 185, 177, 202])
baseball_heights = np.array([72, 74, 78, 80, 76, 82, 79, 73, 81, 77, 75, 80, 78, 74,
83])
baseball_names = np.array([
    "Smith", "Johnson", "Williams", "Brown", "Jones", "Garcia", "Miller", "Davis",
    "Rodriguez", "Martinez",
    "Hernandez", "Lopez", "Gonzalez", "Wilson", "Anderson"
])

# Soccer data
soccer_heights = np.array([180, 175, 178, 182, 177, 185, 172, 179, 181, 176, 174,
183, 180, 178, 177])
soccer_shooting = np.array([0.98, 0.85, 0.76, 0.92, 0.88, 0.95, 0.67, 0.80, 0.91,
0.73, 0.69, 0.87, 0.90, 0.82, 0.79])
soccer_ratings = np.array([7.5, 8.0, 6.8, 7.9, 7.2, 8.3, 6.5, 7.1, 8.1, 7.0, 6.9,
8.2, 7.8, 7.3, 7.0])
soccer_positions = np.array(['A', 'M', 'D', 'A', 'M', 'A', 'D', 'M', 'A', 'D', 'M',
'A', 'M', 'D', 'A'])

# 1. Print out the weight of the first ten baseball players.
print("--- Question 1: First 10 Baseball Weights ---")
# Using array slicing [:10] to grab indices 0 through 9
print(baseball_weights[:10])
print("\n")

# 2. What is the median weight of all baseball players in the data?
print("--- Question 2: Median Baseball Weight ---")
# np.median() calculates the middle value of the dataset
median_weight = np.median(baseball_weights)
print(f"Median Weight: {median_weight} lbs")
print("\n")

# 3. Print out the names of all players with a height greater than 80 inches.
print("--- Question 3: Baseball Players Taller than 80 Inches ---")
# Creating a boolean mask where height > 80, then indexing the names array
tall_players = baseball_names[baseball_heights > 80]
print(tall_players)
print("\n")

# 4. Who is taller on average? Baseball players or soccer players? (Note: Baseball =
inches, Soccer = centimeters)
```

```

print("--- Question 4: Average Height Comparison ---")
# Calculate average baseball height in inches and convert to cm (1 inch = 2.54 cm)
avg_baseball_cm = np.mean(baseball_heights) * 2.54
avg_soccer_cm = np.mean(soccer_heights)

print(f"Average Baseball Player Height: {avg_baseball_cm:.2f} cm")
print(f"Average Soccer Player Height: {avg_soccer_cm:.2f} cm")

if avg_baseball_cm > avg_soccer_cm:
    print("Result: On average, baseball players are taller.")
else:
    print("Result: On average, soccer players are taller.")
print("\n")

# 5. Convert soccer_shooting decimals to whole numbers (e.g., 0.98 becomes 98).
print("--- Question 5: Soccer Shooting Conversion (First 5 Examples) ---")
# Multiply the entire numpy array by 100 and cast as an integer type
soccer_shooting_whole = (soccer_shooting * 100).astype(int)
print(soccer_shooting_whole[:5])
print("\n")

# 6. Calculate the correlation between soccer_ratings and soccer_heights.
print("--- Question 6: Correlation Matrix ---")
# np.corrcoef() returns a correlation matrix. [0, 1] extracts the specific
correlation value.
correlation = np.corrcoef(soccer_ratings, soccer_heights)[0, 1]
print(f"Correlation coefficient: {correlation:.4f}")
print("\n")

# 7. What is the average rating for attacking players ('A')?
print("--- Question 7: Average Rating for Attackers ---")
# Create a boolean filter where position is exactly 'A', then find the mean of their
ratings
attacker_ratings = soccer_ratings[soccer_positions == 'A']
avg_attacker_rating = np.mean(attacker_ratings)
print(f"Average Attacker Rating: {avg_attacker_rating:.2f}")

--- Question 1: First 10 Baseball Weights ---
[180 175 200 190 185 210 195 170 205 188]

--- Question 2: Median Baseball Weight ---
Median Weight: 190.0 lbs

--- Question 3: Baseball Players Taller than 80 Inches ---
['Garcia' 'Rodriguez' 'Anderson']

```

--- Question 4: Average Height Comparison ---

Average Baseball Player Height: 196.77 cm

Average Soccer Player Height: 178.47 cm

Result: On average, baseball players are taller.

--- Question 5: Soccer Shooting Conversion (First 5 Examples) ---

[98 85 76 92 88]

--- Question 6: Correlation Matrix ---

Correlation coefficient: 0.7784